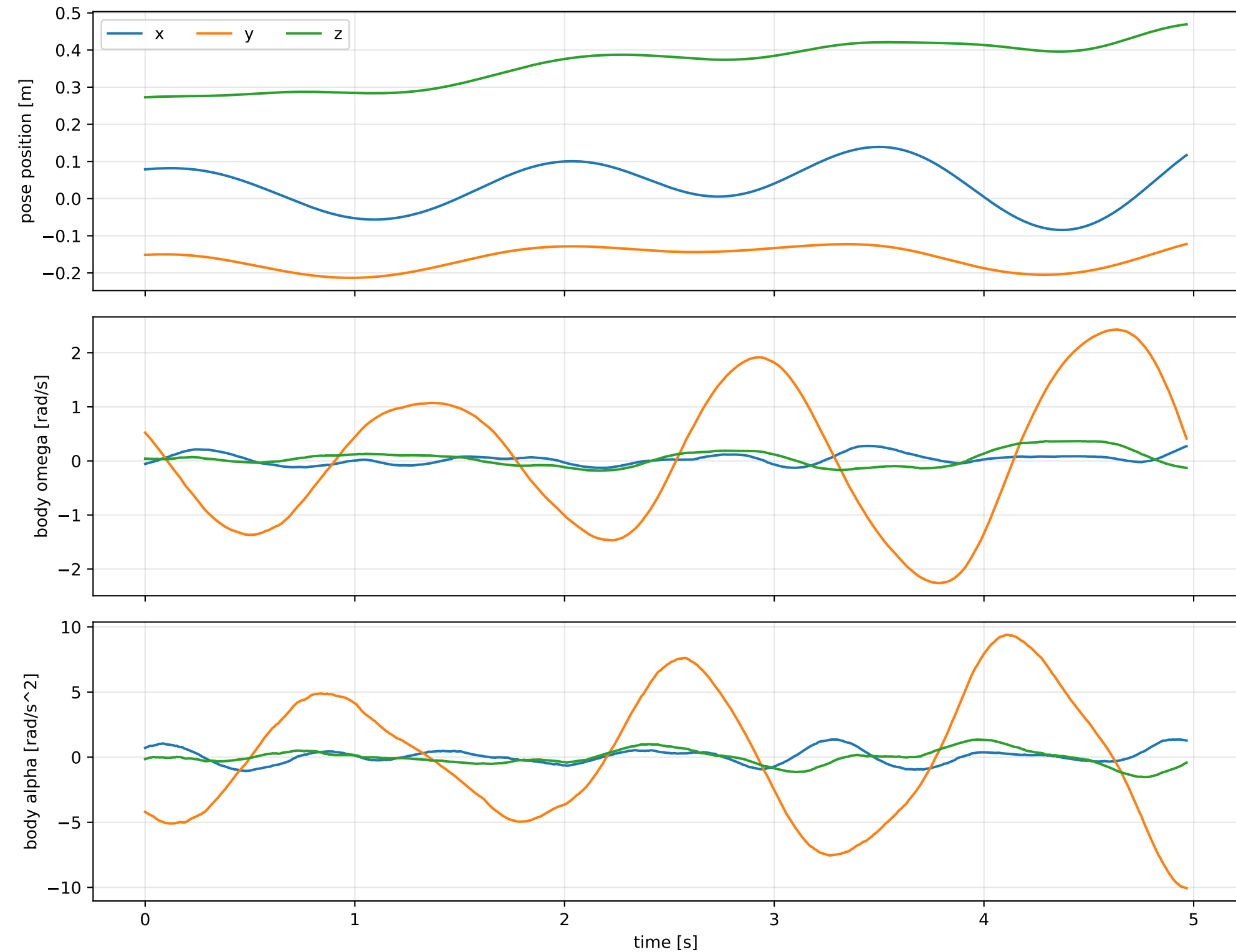
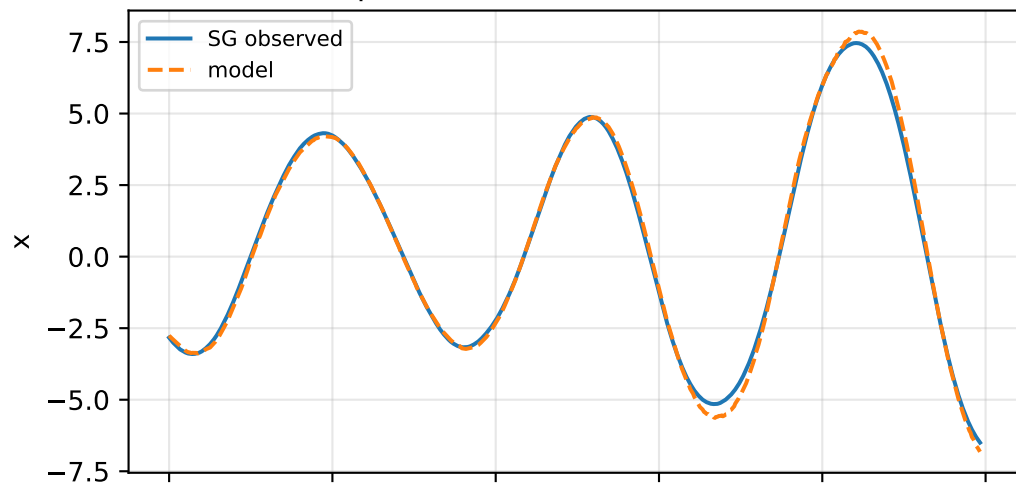


so3_naive_rotation_vector_derivatives: SG trajectory and derived motion

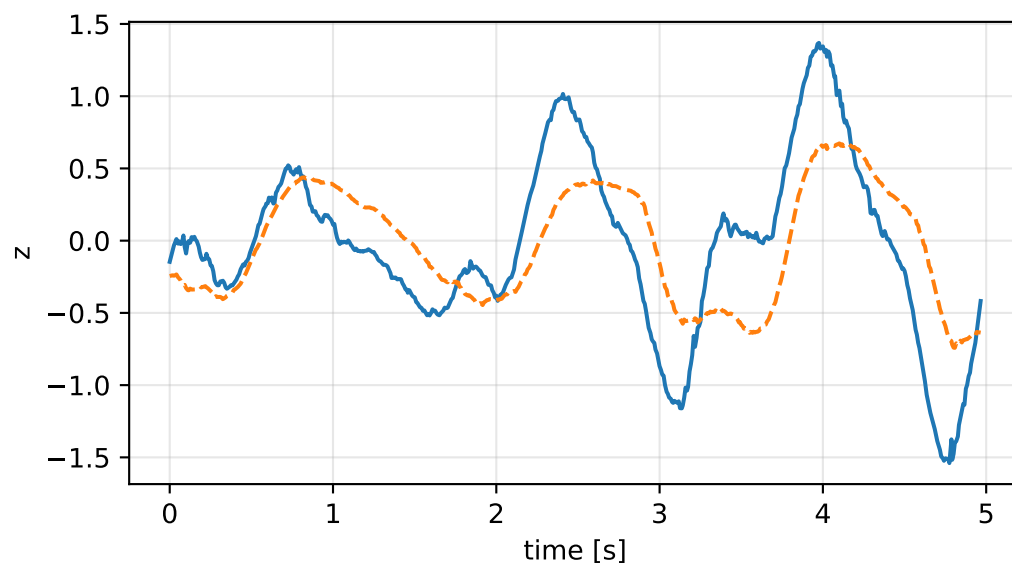
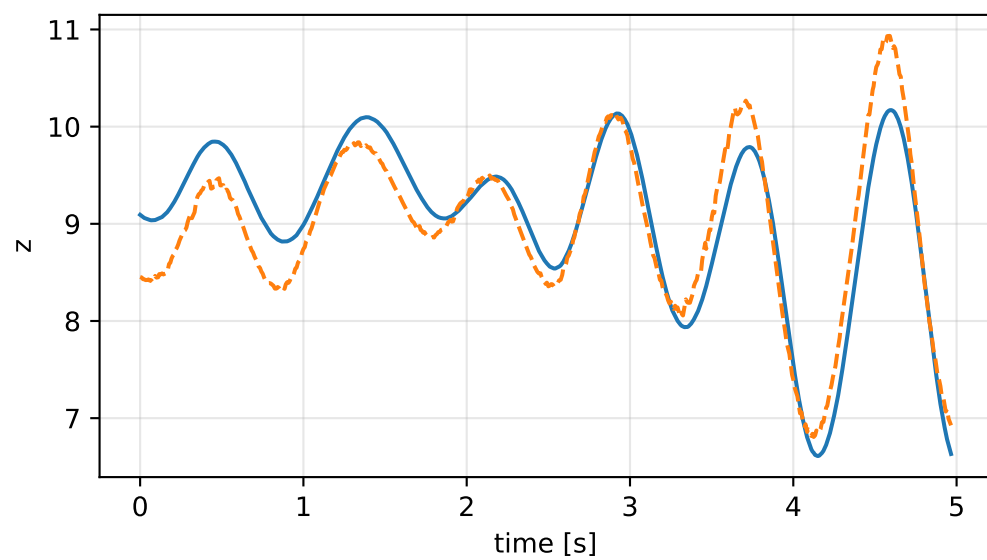
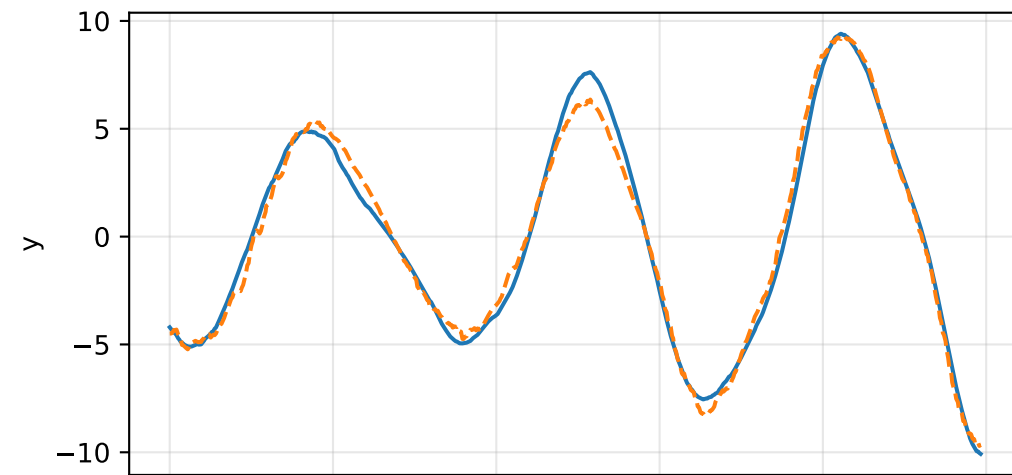
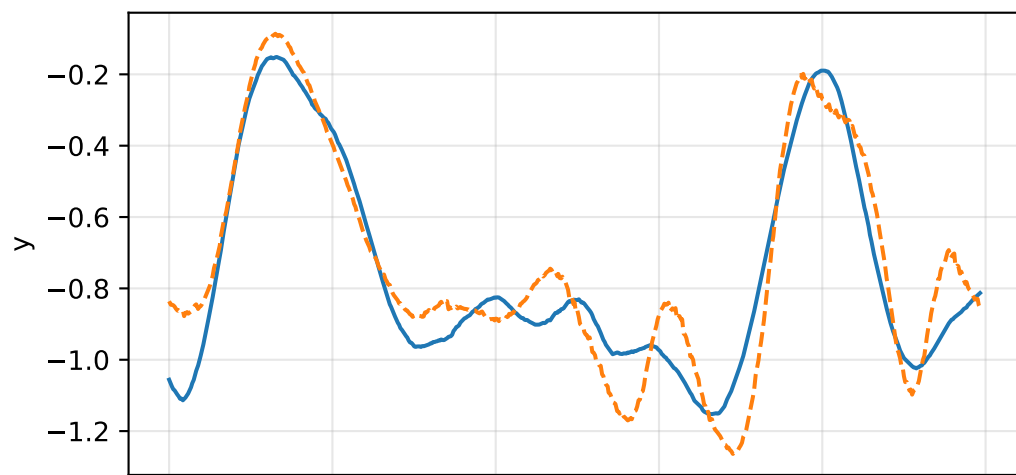
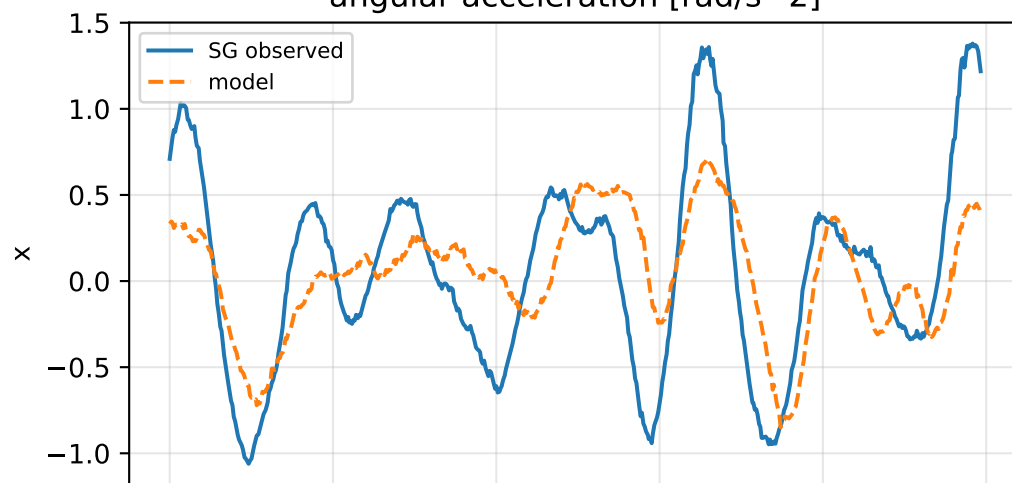


so3_naive_rotation_vector_derivatives: acceleration residual objective

specific acceleration [m/s^2]

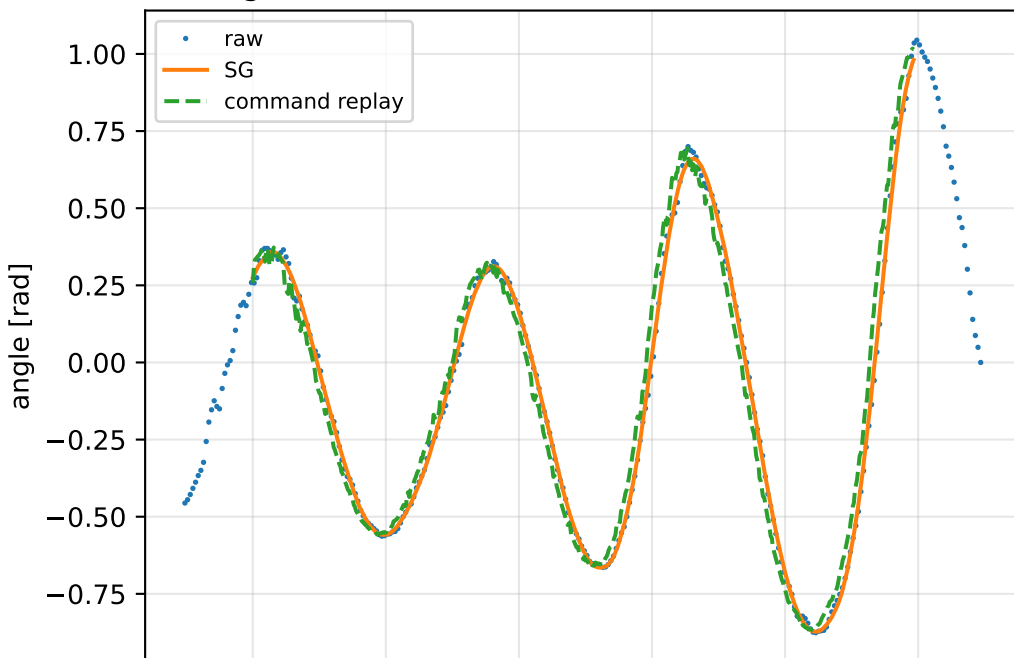


angular acceleration [rad/s^2]

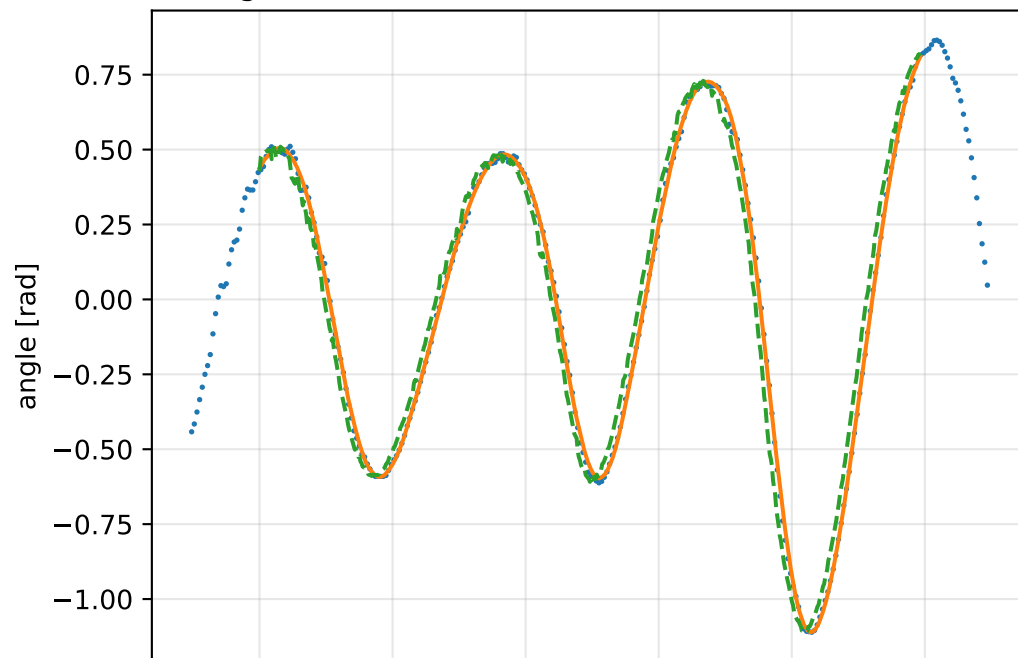


so3_naive_rotation_vector_derivatives: gimbal measurement audit (objective=measured_sg)

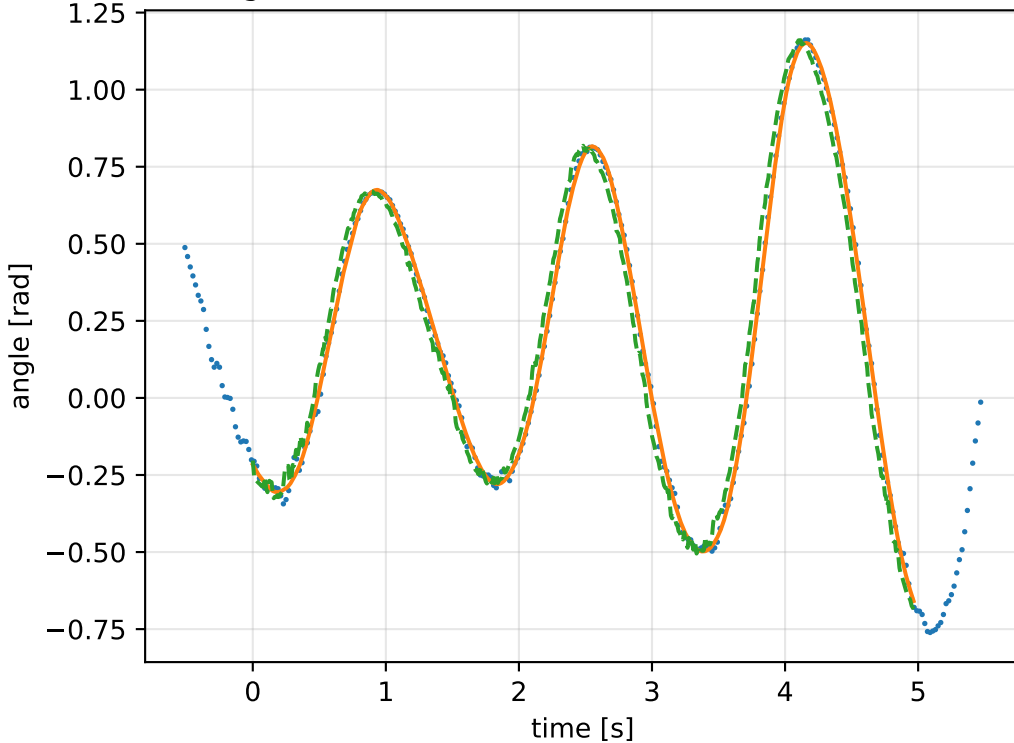
gimbal 1: RMSE=0.0759, max=0.2095 rad



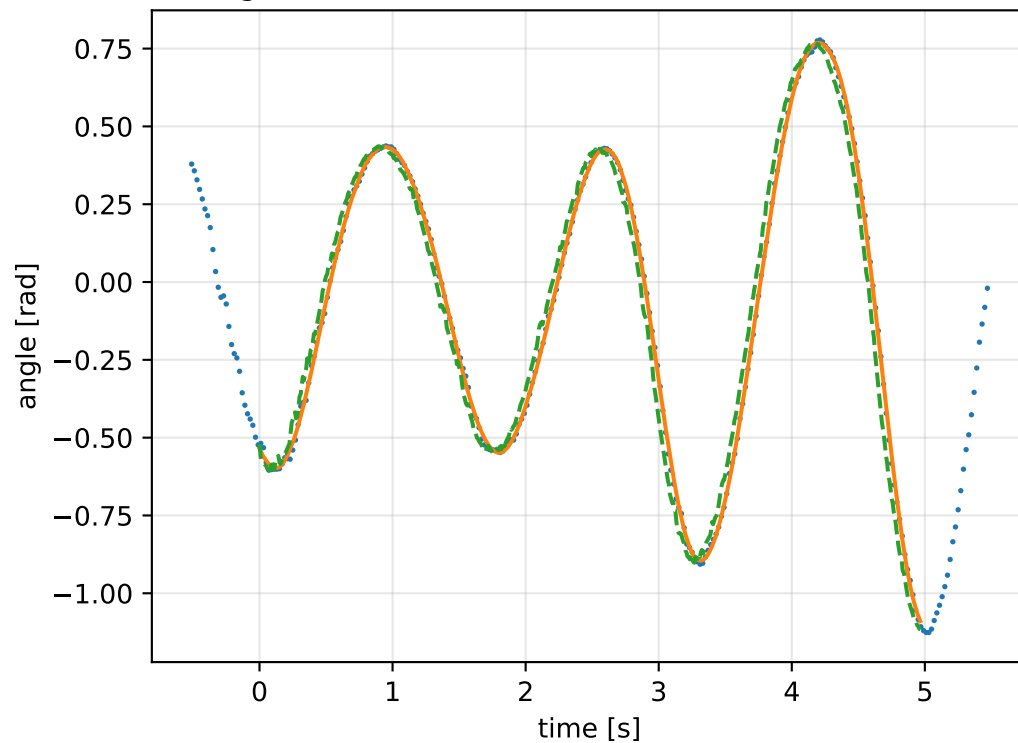
gimbal 2: RMSE=0.08099, max=0.2013 rad



gimbal 3: RMSE=0.07393, max=0.172 rad

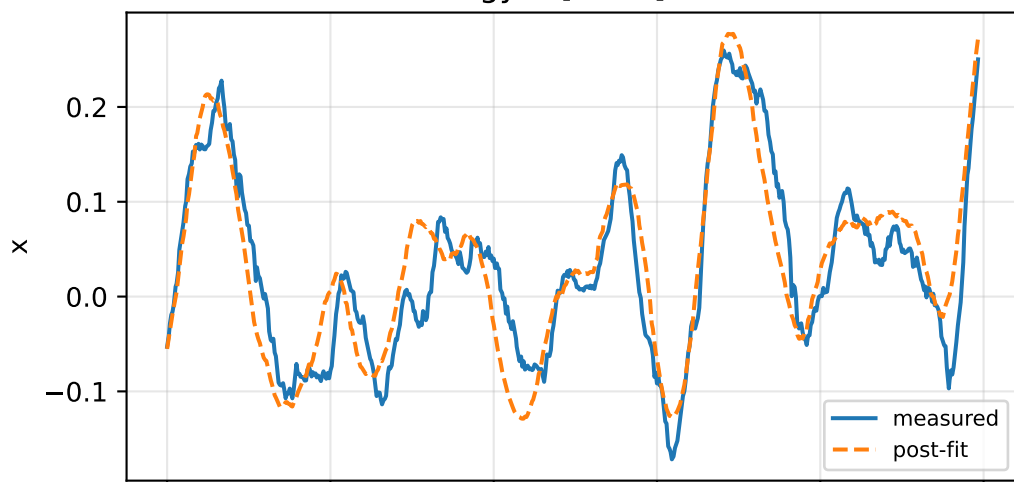


gimbal 4: RMSE=0.07131, max=0.1737 rad

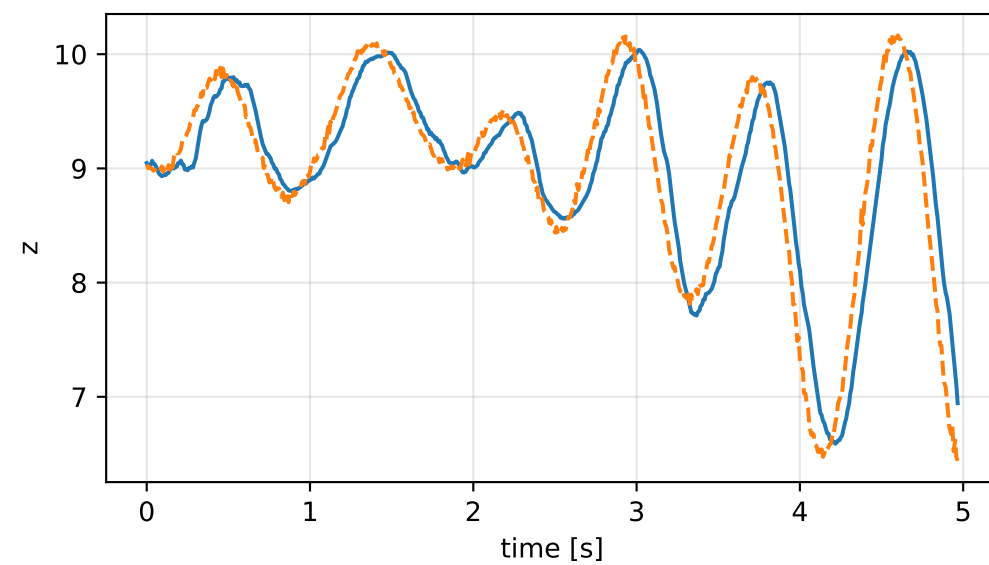
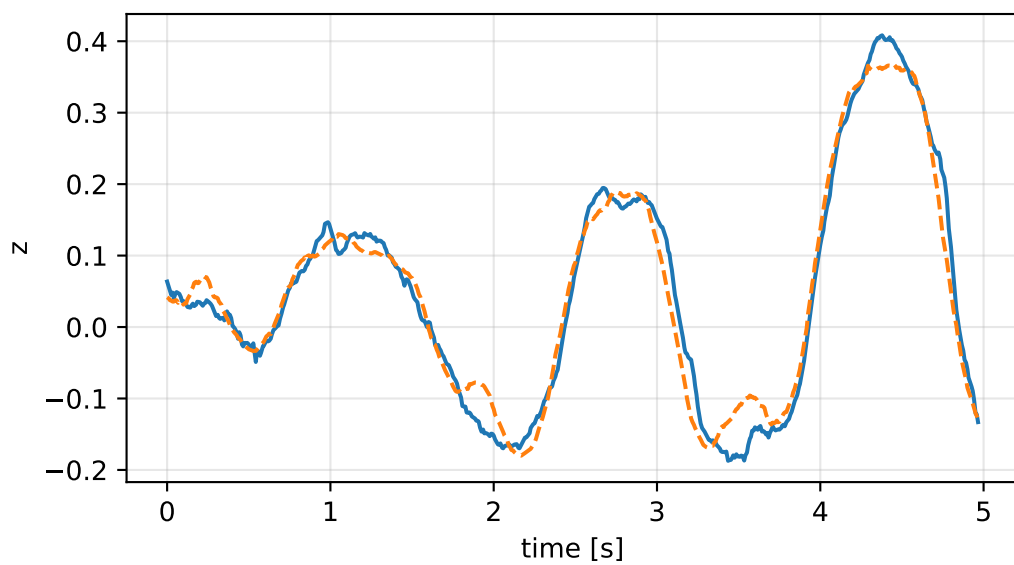
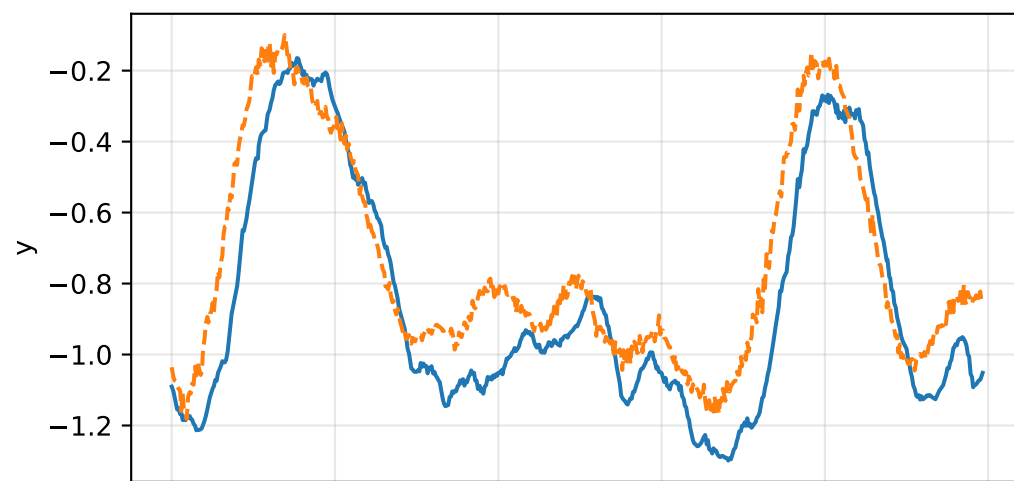
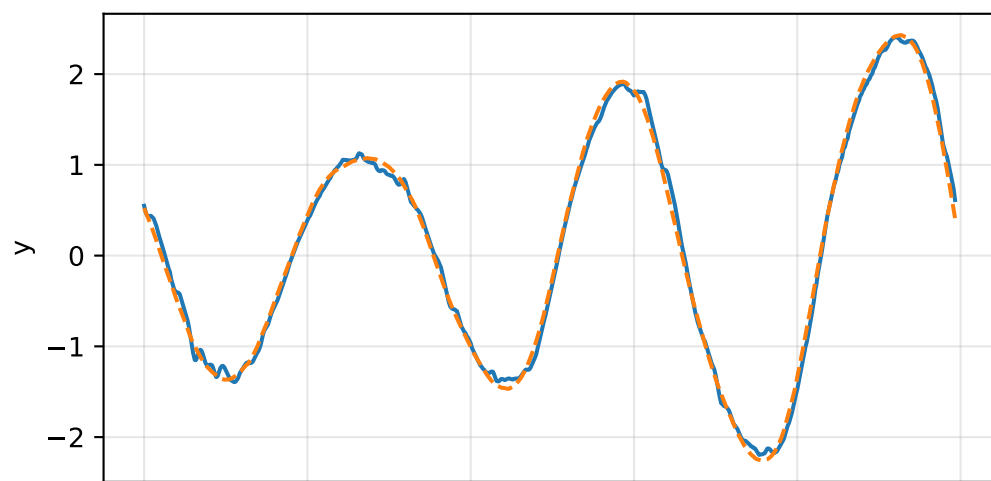
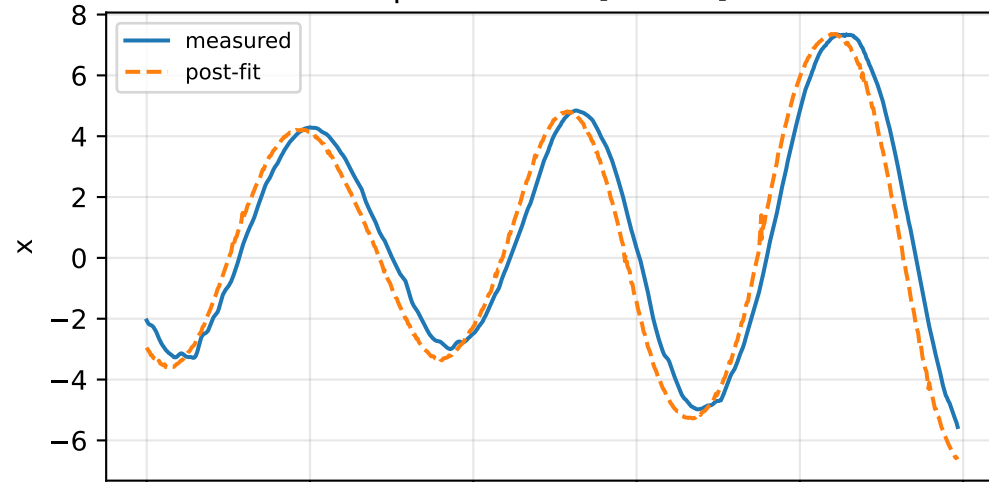


so3_naive_rotation_vector_derivatives: IMU comparison (diagnostic only)

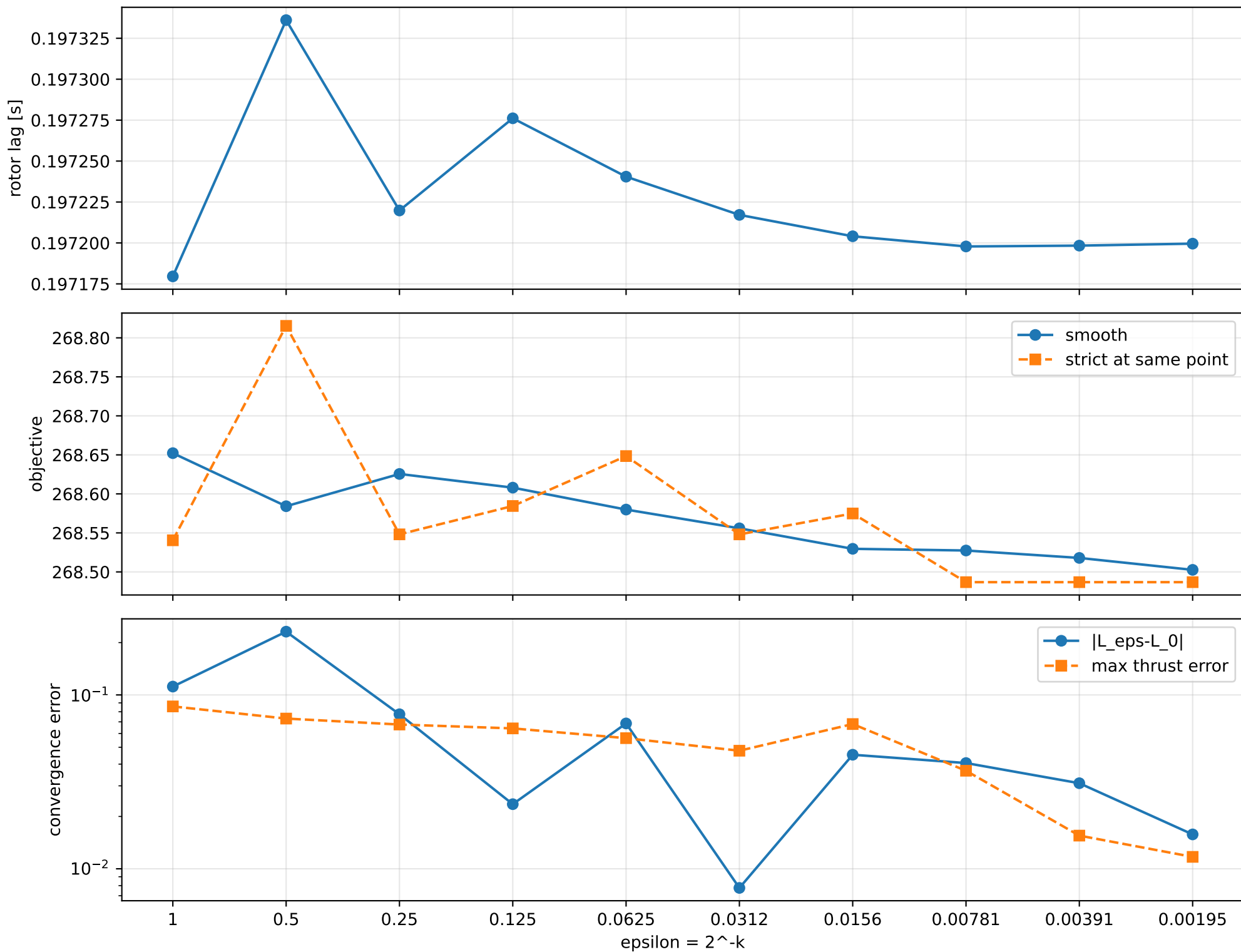
gyro [rad/s]



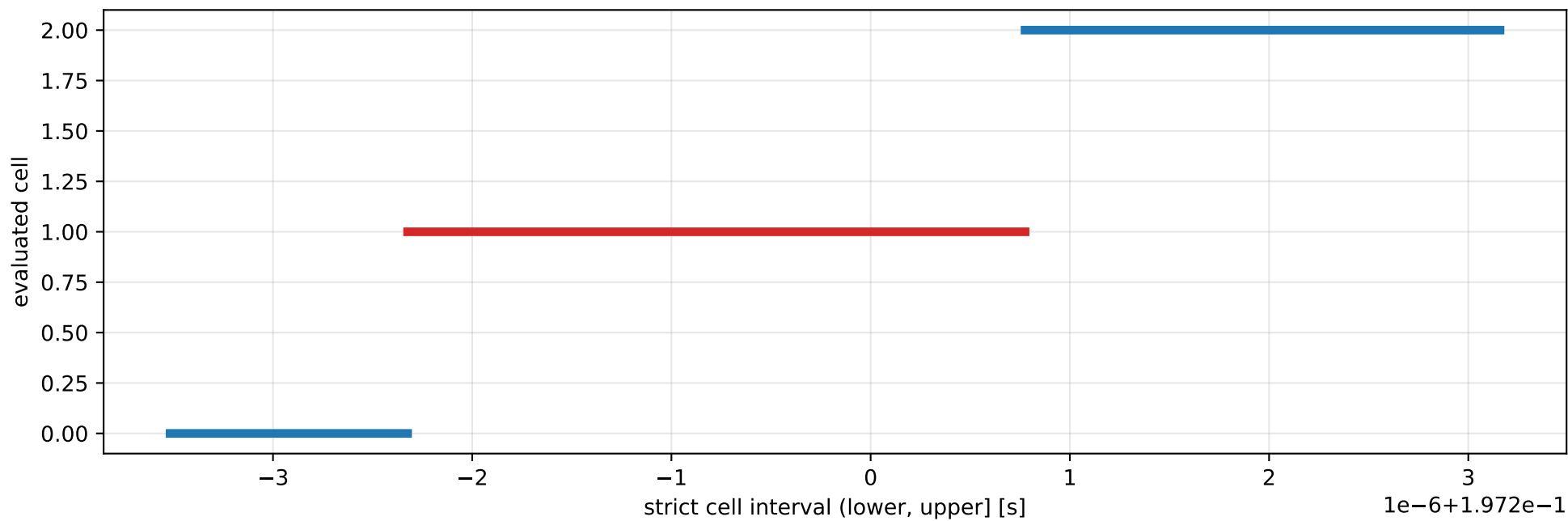
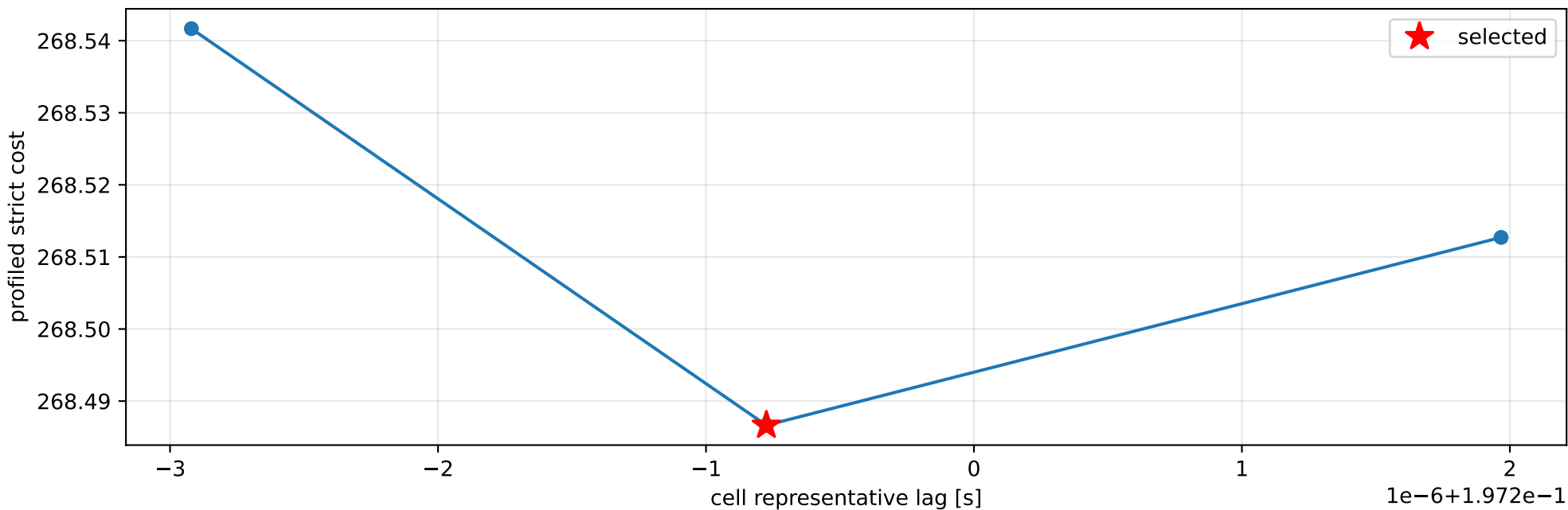
specific force [m/s²]



so3_naive_rotation_vector_derivatives: smooth-to-strict continuation



so3_naive_rotation_vector_derivatives: exact strict-ZOH lag cells



selected (0.197197676, 0.197200775] s; final smooth 0.197199569 s; data boundary=False

so3_naive_rotation_vector_derivatives: parameter estimate

Case: so3_naive_rotation_vector_derivatives

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 3.28648298

inertia [kg m²]:

```
[[ 3.35628508e-01 -2.68557159e-04 -5.51525775e-02]
 [-2.68557159e-04  2.54292080e-01 -2.12817834e-04]
 [-5.51525775e-02 -2.12817834e-04  5.64731611e-01]]
```

CoG body [m]: [0.01767736 -0.00398435 -0.04439558]

force effectiveness: [1.0996542 1.41692562 1.23860061 1.02665474]

scale-free J/m [m²]:

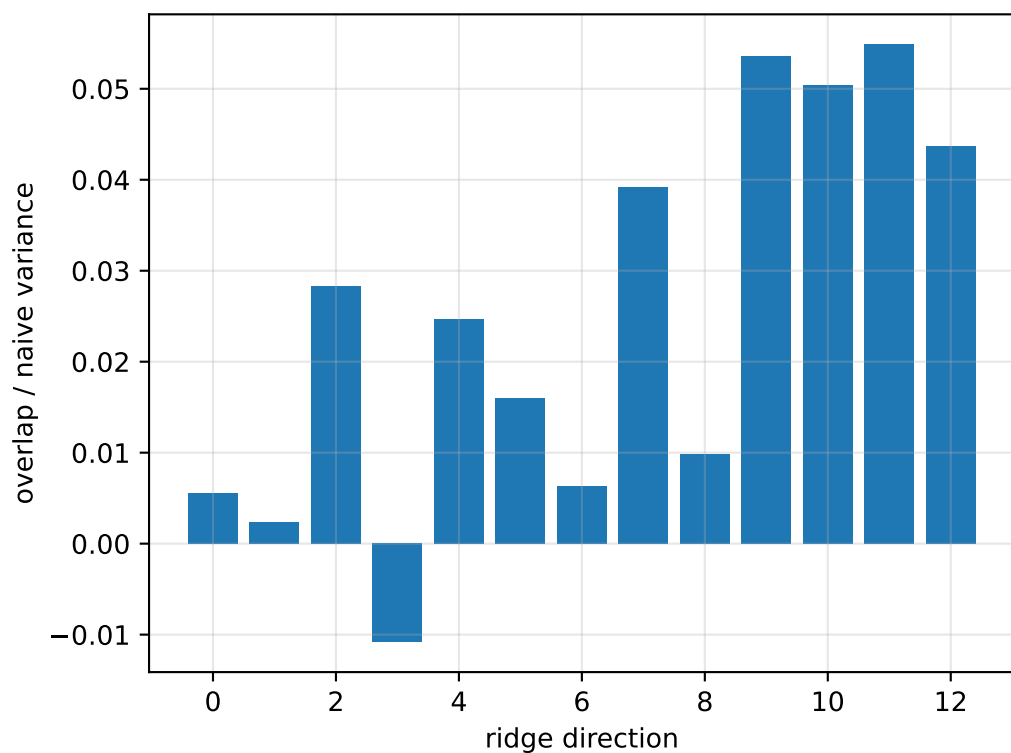
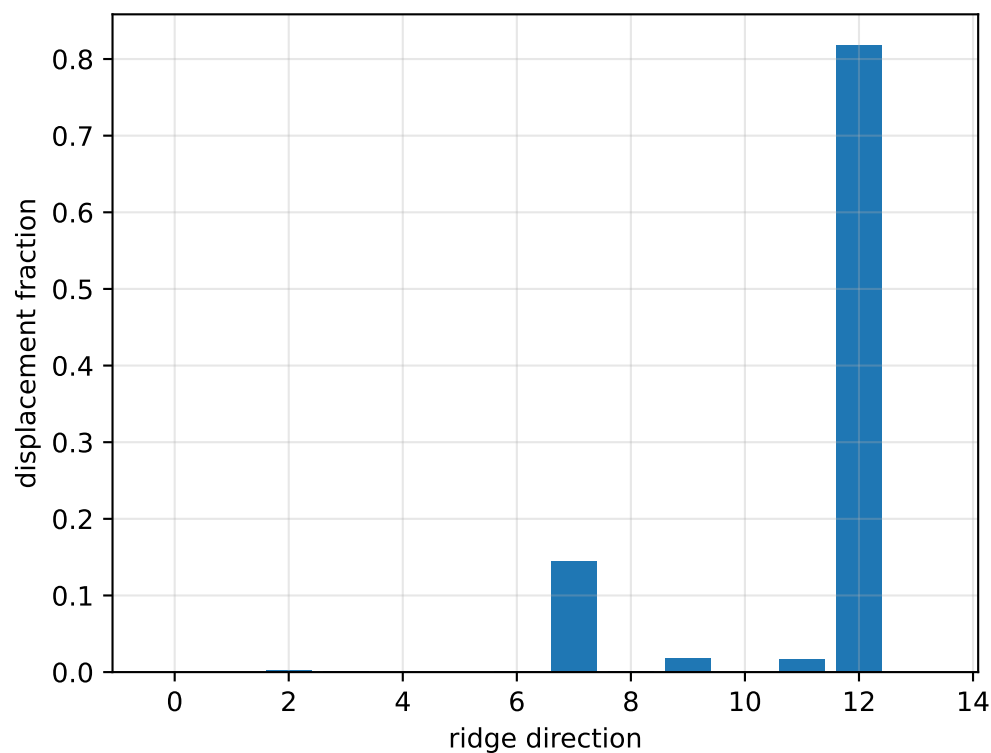
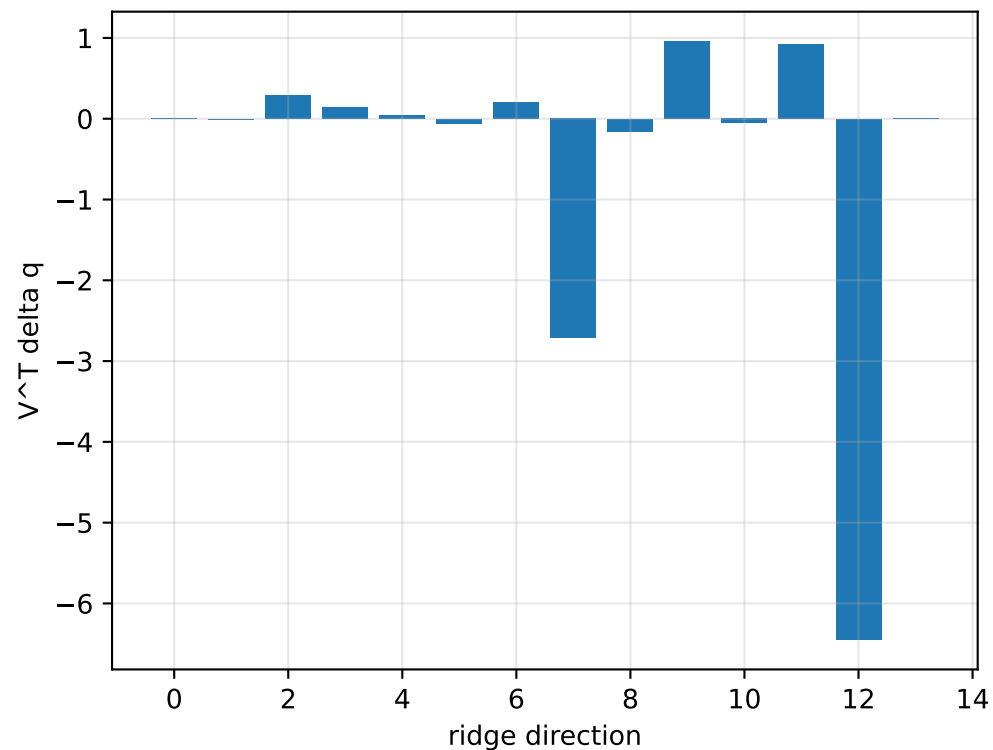
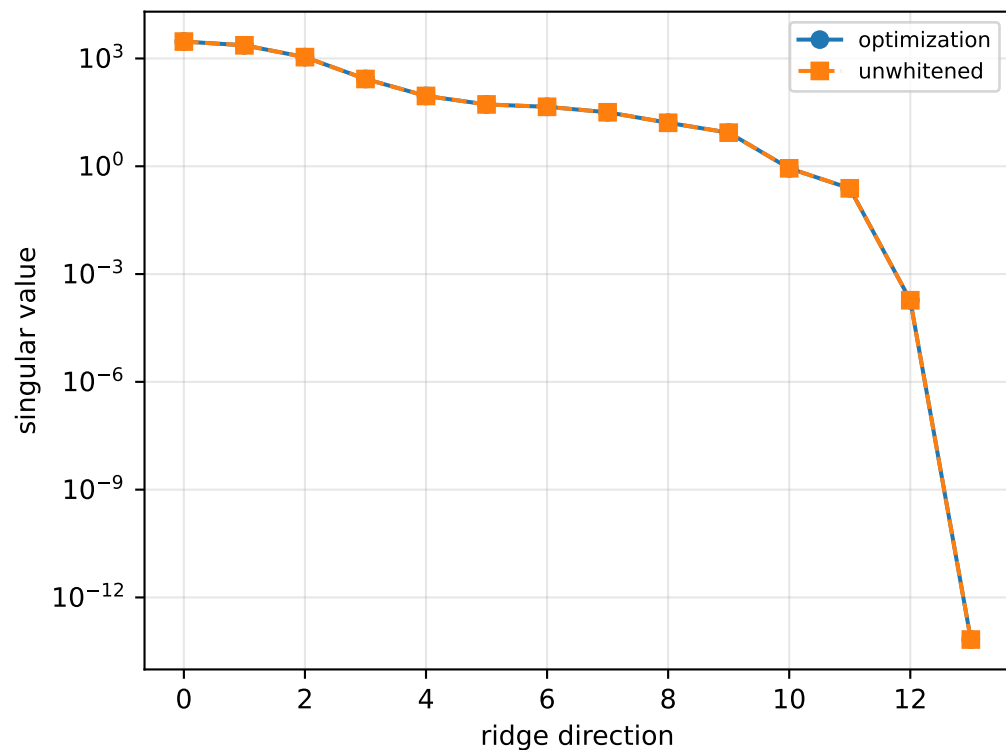
```
[[ 1.02123915e-01 -8.17156701e-05 -1.67816410e-02]
 [-8.17156701e-05  7.73751398e-02 -6.47554954e-05]
 [-1.67816410e-02 -6.47554954e-05  1.71834637e-01]]
```

scale-free f/m: [0.33459909 0.43113737 0.37687723 0.31238705]

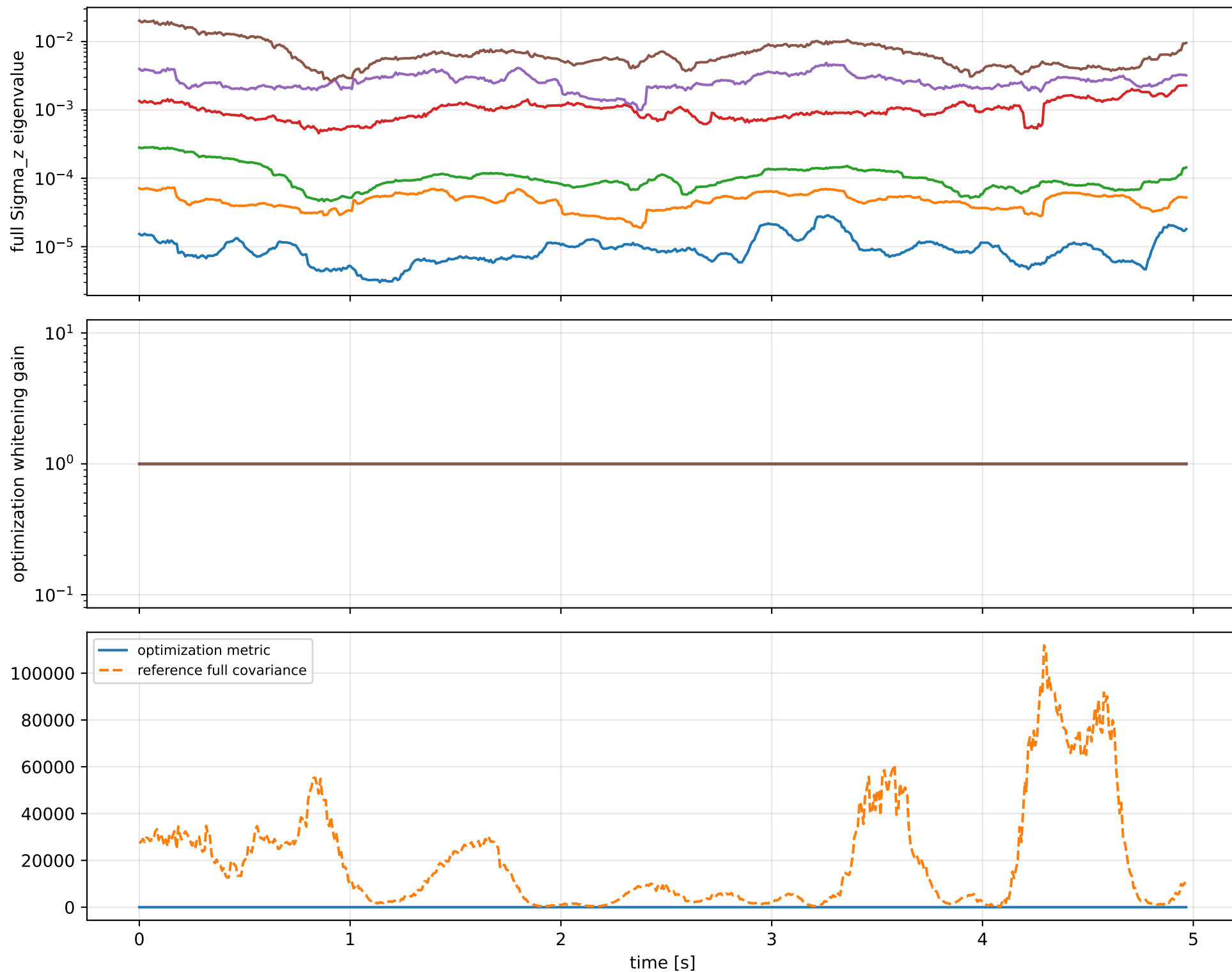
rotor lag cell: (0.197197676, 0.197200775] s

representative: 0.197199225 s

so3_naive_rotation_vector_derivatives: ridge spectrum (rank 13, nullity 1, ||v||=9.21e-14)



so3_naive_rotation_vector_derivatives: optimization vs reference covariance



so3_naive_rotation_vector_derivatives: wrench / closure (force max $1.15\text{e-}14$, torque max $9.51\text{e-}16$)

